

# TCS Toolkit

Hector Jimenez

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## 1 Asymptotics and Probability

### 1.1 Big $O$ and friends

Recall that  $f(x) = O(g(x))$  (as  $x \rightarrow \infty$ )  $\iff \exists c, x_0$  s.t.  $|f(x)| \leq c \cdot g(x)$ ,  $\forall x \geq x_0$ . **Example:**  $\sum_{i=1}^n i = \frac{n(n+1)}{2} = \frac{n^2}{2} + \frac{n}{2} = O(n^2) = \frac{n^2}{2} + O(n)$ .

**Definition 1.1** (Harmonic Number). Let  $n \in \mathbf{N}$ , then we define  $H_n = 1 + 1/2 + 1/3 + 1/4 + \dots + 1/n$ .

We can get an **upper bound** and a **lower bound** for harmonic number in the following way:

$$\begin{aligned} H_n &= 1 + 1/2 + 1/3 + 1/4 + \dots + 1/n \leq \\ &1 + 1/2 + 1/2 + 1/4 + 1/4 + 1/4 + 1/4 + 1/8 + \dots \leq \\ &\quad \lfloor \log_2 n \rfloor + 1. \end{aligned}$$

$$\begin{aligned} H_n &= 1 + 1/2 + 1/3 + 1/4 + \dots + 1/n \geq \\ &1 + 1/2 + 1/4 + 1/4 + 1/8 + 1/8 + 1/8 + 1/8 + 1/16 \dots \geq \\ &\quad 1/2 \lceil \log_2 n \rceil + 1. \end{aligned}$$

Since that  $H_n \leq O(\log n)$  and  $H_n \geq \Omega(\log n)$ , we can conclude that  $H_n = \Theta(\log n)$ .

**Definition 1.2** ( $f(x) \sim g(x)$ ). We say that  $f(x) \sim g(x)$  as  $x \rightarrow \infty$  if  $\frac{f(x)}{g(x)} \rightarrow 1$ .

We recall the definition of a Taylor Series:

**Theorem 1.1** (Taylor Series). If  $f(x)$  is analytic at  $x = a$ , then it can be represented by its Taylor series:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n,$$

which converges to  $f(x)$  for all  $x$  within the radius of convergence.

**Definition 1.3** ( $e^x$ ). We define the function  $e^x = \sum_{i=0}^{\infty} \frac{x^i}{i!}$ ,  $\forall x \in \mathbf{R}$ .

For example, the function  $\ln 1+x = \sum_{n=0}^{\infty} (-1)^{n-1} \frac{x^n}{n!}$ , for  $|x| \leq 1$ . Additionally, we can use this equivalence to approximate  $\ln 1+x = \sum_{n=0}^{\infty} (-1)^{n-1} \frac{x^n}{n!} \sim x$  when  $x \rightarrow 0^+$ . We can also exponent this and get:  $\exp x \sim 1+x$ .

**Definition 1.4** (Central Binomial Coefficient). We define the central binomial coefficient as  $C_n := \binom{n}{n/2}$ . We notice that the probability of getting half of  $n$  flipped coins is  $\frac{C_n}{2^n}$ .

**Proposition 1.2.**  $C_n = \Theta\left(\frac{2^n}{\sqrt{n}}\right)$ .

*Proof.* We can bound above and below the expression. Lets do a simple example to see this.  $C_{10} = \frac{10 \cdot 9 \cdot 8 \cdot \dots \cdot 2 \cdot 1}{5 \cdot 4 \cdot \dots \cdot 1 \cdot 5 \cdot 4 \cdot \dots \cdot 1}$ . Then we divide by  $2^{10}$  and get:  $C_{10}/2^{10} = \frac{10 \cdot 9 \cdot 8 \cdot \dots \cdot 2 \cdot 1}{10 \cdot 8 \cdot \dots \cdot 2 \cdot 10 \cdot 8 \cdot \dots \cdot 2}$ . Furthermore we can potentiate to  $(\cdot)^2$  and get:  $\left(\frac{C_n}{2^n}\right)^2 = \frac{10 \cdot 10 \cdot 9 \cdot 9 \cdot 8 \cdot 8 \cdot \dots \cdot 2 \cdot 2 \cdot 1}{10 \cdot 10 \cdot 10 \cdot 10 \cdot 8 \cdot 8 \cdot 8 \cdot 8 \cdot \dots \cdot 2 \cdot 2 \cdot 2 \cdot 2} = \frac{9 \cdot 9 \cdot 7 \cdot 7 \cdot 5 \cdot 5 \cdot 3 \cdot 3}{10 \cdot 10 \cdot 8 \cdot 8 \cdot 6 \cdot 6 \cdot 4 \cdot 4 \cdot 2 \cdot 2}$ . From this we can get an upper bound:  $\left(\frac{C_n}{2^n}\right)^2 \leq \frac{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{11 \cdot 10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2} = \frac{1}{11} = \frac{1}{n+1}$ . Also we can get a lower bound:  $\left(\frac{C_n}{2^n}\right)^2 \geq \frac{9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{10 \cdot 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = \frac{1}{10} = \frac{1}{n}$ .  $\square$

## 1.2 Sum of Random Variables, Gaussian and Central Limit Theorem

We work with the following setup: let  $X_1, X_2, \dots, X_n$  be i.i.d random variables with  $\mathbf{P}[X_i = 1] = p$  and  $\mathbf{P}[X_i = 0] = 1 - p = q$ . Let  $S_n = X_1 + X_2 + \dots + X_n$ , then  $\mathbf{E}(S_n) = \mathbf{E}(X_1) + \dots + \mathbf{E}(X_n) = pn$ .

## 2 Circuits

**Definition 2.1** (Circuit). A circuit is a Directed Acyclic Graphs (DAGs) that computes Boolean Functions  $f : \{0, 1\}^n \rightarrow \{0, 1\}$ . The nodes of a circuit are called gates. A circuit can be seen as a tree, where the leafs are input gates that receives an input  $x = (x_1, x_2, \dots, x_n)$  and the root generates an output bit.

**Definition 2.2** (Size and Depth of Circuit). The Size and Depth of a circuit  $C$  are defined as the number of non-input gates and the longest input-output path, respectively.

**Example 2.1.** Some common basis are the following:  $\mathcal{B}_1 = \{\neg, \text{binary } \vee, \text{binary } \wedge\}$ ,  $\mathcal{B}_2 = \{\text{all } g : \{0, 1\}^2 \rightarrow \{0, 1\}\}$ ,  $\mathcal{B}_3 = \{\neg, \text{all fan-in } \vee, \text{all fan-in } \wedge\}$ . Notice that  $\mathcal{B}_3$  allow for constant depth computations. For example, we can add two  $n$ -bits numbers in constant depth. Instead for  $\mathcal{B}_1$ , we need at least a depth of  $\log n$  to look all bits.

**Example 2.2.** We study the function **AND**:  $\{0, 1\}^n \rightarrow \{0, 1\}$ . Notice that with  $\mathcal{B}_3$  we can generate a circuit with size 1. Instead with  $\mathcal{B}_2$  we can generate a circuit with size  $O(n)$  and depth  $O(\log n)$ .

**Example 2.3.** We study the **Palindrome** function  $p : \{0, 1\}^{2^n} \rightarrow \{0, 1\}$ . With  $\mathcal{B}_3$  we can generate a circuit with depth 2, by generating an "equal" gate.

**Definition 2.3** (Basis). We call  $\mathcal{B}(C)$  the basis of a circuit  $C$ , which is a set of allowed operations in gates of a circuit  $C$ .

We have talk about constant size inputs of size  $n$ , this mean that for each input size we need to generate a different circuit. In order to accept a certain language  $L$ , we define a **circuit family**  $(C_n)_{n=1}^\infty$ , where the circuit  $C_n$  should accepts all words in  $L \cap \{0, 1\}^n$ . Now, we define some families of languages that are categorized by the size of circuit families (with  $\mathcal{B}_3$ ) that accept them.

**Definition 2.4** (Family  $AC^0$ ). The collection of languages decided by a circuit family of depth  $O(1)$  and size  $\text{poly}(n)$ .

**Definition 2.5** (Family  $NC$ ). The collection of languages decided by a circuit family of depth  $\text{polylog}(n)$  and size  $\text{poly}(n)$ .

There are another notations about language decidable by circuits.

**Definition 2.6** (P-Poly). The family of languages decidable by  $\text{poly}(n)$ -size circuits size.

A cool proposition about circuits.

**Proposition 2.1** (Has 86). There exists a language  $L \in P$  with no  $\text{poly}$ -size, constant-depth circuits.

**Proposition 2.2** (CLIQUE). CLIQUE requires exponential size AND/OR circuits.